PHARMACY INVENTORY SYSTEM

GROUP 10

Before solutions can be implemented, the problem must be fully understood. This initial phase involves comprehensive assessment of existing inventory workflows, data collection on stockouts and expired medications, and stakeholder engagement, establishing the blueprint for meaningful improvement.

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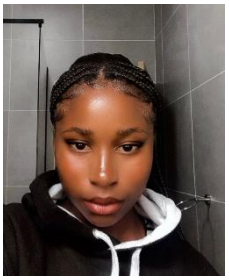
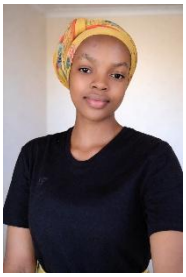
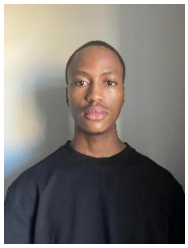
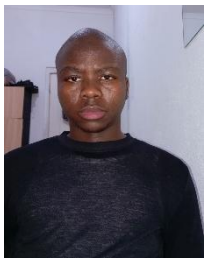
 Mthimunye Sakhile (Group Leader)	 Sedze Tanaka Bruce	 Mashao Keabetswe Beyonce	 Shibambo Karabo
 Mbele Paul Mpho	 Konyana Lethabo	 Lamola Charmaine	 Mokgohloa Mpho

Comprehensive Company Analysis, Problems, and Constraints

1. COMPANY SITUATION ANALYSIS

1.1. Company Mission

To provide a centralized, efficient, and accurate inventory management system that ensures the right pharmaceutical products are available at the right time, with the right quantity and within expiry dates, to support safe and effective patient care.

1.2. Strategic Objectives

- Efficient maintenance of stock accuracy and tracing or recording every stock movement.
- To prevent, monitor, and control medicine expiry dates.
- To store medicines in correct environmental conditions.
- Ensure that stock running-out alerts are always automated to prevent stockouts of urgent medicines.
- Prepare effectual orders.
- To generate and maintain accurate auditable dispensing records for every transaction.
- Ensuring full compliance with applicable pharmaceutical legislation along with the right technological tools to maintain data integrity.
- Additional Objectives from Analysis:
    - Streamlining day-to-day processes through digitized systems to reduce human errors and enhance productivity.
    - Guaranteeing patient satisfaction through prompt service, accurate prescription fulfilment, and reliable information.

1.3. Company Operations (Day-to-Day Functions)

The pharmacy operates daily with the following core operational activities:

Operational Area	Daily Activities
Stock Receiving and Intake Management	Have a Stock Controller physically manage delivered stock and verify if it is the correct order. Record the medicine batch in the system with batch number, quantity received, expiry dates, manufacturing data, and unit cost. Update the system immediately after stock has been received.
Stock Placement	Received products are placed on shelves according to category (prescription, over the counter, scheduled substances) and storage requirements (room temperature, cold chain). System locations are updated to reflect where each product is stored.
Stock Storage and Monitoring	Provide appropriate, safe, and secure storage for medicines according to their temperature. Attention to the cleanliness of storage facilities. The system continuously tracks quantities on hand per batch, flags stock whose expiry dates are within 30 or 60 days, and generates alerts when stock is about to run out. Staff perform periodic checks of fast-moving items.
Prescription Processing	When a prescription is received, the Dispensing Pharmacist processes it. The system checks for drug interactions, allergies, and repeats remaining. Stock availability is verified before dispensing. Prescription details are deliberately recorded in the system.
Dispensing	Prioritize medicines that are close to expiring when dispensing. Ensure the patient gets the right medicine. Record the transactions immediately after dispensing. The pharmacist and patient's details should also be captured in the system.
Obtainment and Supplier Management	Keep track of medicine stock availability. Generate alerts if stock is running out. Ensure efficient communication between the obtained of the pharmacy and the relevant supplier. Track the purchase order with the expected delivery date.

End-of-Day Reconciliation	Staff verify stock counts for controlled substances sales totals are reconciled with cash and card payments; daily reports are generated for management review.

1.4. Company Organizational Structure

The pharmacy's organizational structure is designed to support efficient workflow and accountability across all operational areas.

Role	Key Responsibilities
Chief Executive Officer	Responsible for company's directions.
Operations Manager	Oversee how the pharmacy operates daily.
Pharmacy Manager/ Head Pharmacist	Overall supervision of pharmacy services, ensures regulatory compliance, handles controlled substances, final verification of prescriptions, and authorizes procurement.
Responsible Pharmacist	Daily pharmacy operations, prescription verification, patient counselling, staff supervision
Dispensing Pharmacist	Processes patient prescriptions and verifies if the medicines are coherent.
Inventory Manager	Oversees stock receiving, placement, and ordering, manages expiry tracking; coordinates stock takes.
Stock Controller	Receive the delivered stock in the pharmacy, monitor the storage, communicate with the suppliers, updates stock records, and processes returns to suppliers.
Pharmacy Assistants	Responsible for day-to-day customer service, assist with prescription intake, prepare medications for pharmacist verification, and support the manager in routine operational tasks.
Sales Assistants	Process sales transactions, assist customers with over-the-counter products, maintain shelf displays.
Database Administrator	Maintains the database and ensures all digital operations function smoothly and securely.
Finance Manager	Manages the budgets and maintains financial transaction records.

2. PROBLEMS AND CONSTRAINTS

2.1. Identified Problems

The pharmacy faces several challenges that delays top performance:

Problem ID	Problem Description	Impact
P1	Manual Stock Tracking: Current system relies on manual counts and spreadsheets, leading to inaccurate stock levels and frequent stockouts.	Lost sales, customer dissatisfaction, and inefficient staff time.
P2	Expiry Date Mismanagement: No automated system to track expiry dates; expired products are occasionally found on shelves.	Financial loss (unsellable stock), potential patient safety risk, and regulatory non-compliance.
P3	Prescription Stock Verification: Staff cannot quickly verify if the prescribed medication is in stock before processing the prescription.	Delays in dispensing, frustrated patients, and inefficient workflow.
P4	Reordering Inefficiency: Purchase orders are generated manually based on visual inspection rather than data-driven reorder points.	Overstocking of slow-moving items, understocking of fast-moving items, and cash flow issues.
P5	Controlled Substance Tracking: Schedule 5 and 6 substances require strict audit trails; current system lacks automated logging and reporting.	Compliance risk, time-consuming manual registers, and potential for errors in reporting.
P6	Supplier Order Tracking: No central system to track which products were ordered from which supplier, order dates, and delivery status.	Difficulty resolving delivery discrepancies and managing supplier performance.
P7	Limited Reporting Capabilities: Current reporting mechanisms are labor-intensive and time-consuming.	Limits the pharmacy's ability to make timely, data-driven decisions.

2.2. System Constraints

The project and system development must operate within the following constraints:

Constraint ID	Constraint Description	Implication

C1	Regulatory Compliance: The system must comply with South African Pharmacy Council (SAPC) regulations, the Medicines and Related Substances Act, and Good Pharmacy Practice rules.	Controlled substances must have full audit trails; prescription records must be retained for a minimum period.
C2	POPIA Compliance: Patient information must be protected according to the Protection of Personal Information Act (POPIA).	Access controls, encryption, and audit logging are mandatory.
C3	Data Integrity Constraint: The system must enforce data integrity to ensure accuracy and reliability.	The database must be designed with validation rules and referential integrity to prevent incorrect or orphaned data.
C4	Budget Constraints: This is an academic project with no commercial budget for licensing or cloud hosting.	System will be developed using Oracle Database and will be a prototype, not a live production system.
C5	Time Constraints: Project must be completed within the academic semester.	Scope must be manageable; not all pharmacy features can be fully implemented.
C6	Integration Limitations: The system will not integrate with existing point-of-sale hardware or external medical aid systems.	Some processes (e.g., automatic medical aid claims) are out of scope.

3. Database System Specifications

3.1 Objectives to Address Identified Problems

The database system is designed to transition the pharmacy from a manual, error base environment to an automated and efficient digital system. It addresses the following key problems:

- Inventory Accuracy (P1): Eliminate manual stock tracking by implementing real-time monitoring of medication inventory.

- Expiration Control (P2): Improve management of expiry dates by automatically tracking and notifying staff of medications nearing expiration.
- Prescription Efficiency (P3): Enable fast prescription verification by allowing staff to quickly check medication availability before dispensing.
- Automated Reordering (P4): Prevent stock shortages by setting reorder levels and automatically generating purchase orders.
- Regulatory Compliance (P5): Ensure proper tracking and documentation of Schedule 5 and Schedule 6 medications in accordance with legal requirements.
- Supplier Management (P6): Maintain centralized records of suppliers, orders, and deliveries.
- Reporting (P7): Provide real-time reports on stock levels, transactions, and prescriptions to support decision-making.

3.2 Information Required from the Database

To support pharmacy operations, the database must store and manage the following categories of data:

Inventory Information

- Medication details (name, category, schedule, dosage form)
- Batch information (batch number, quantity, cost, manufacturing date, expiration date)
- Storage details (location and required conditions)

Prescription & Patient Information

- Patient details (name, contact information, allergies)
- Prescription details (doctor, date issued)
- Dispensing records (medication, quantity, pharmacist responsible)

Supplier & Orders Information

- Supplier details (name, contact information, address)
- Purchase orders (order date, delivery date, status)
- Quantities of medications ordered.

Stock Movement Information

- Stock transactions (stock in/out)
- Dates and quantities of movement

- Staff responsible for each transaction

Staff Information

- Staff details (name, role, contact information)
- Roles such as Pharmacist, Stock Controller, and Finance Manager for access control

3.3 Scope

The system will manage the full lifecycle of pharmaceutical inventory within the pharmacy. Its primary functions include:

- Product Management: Add, update, and deactivate products with all relevant attributes
- Stock Management: Track stock levels by batch and expiry date; record stock receipts; record stock adjustments (damage, returns)
- Prescription Processing: Capture prescriptions; verify stock availability; record dispensing; update stock quantities
- Automated Reordering: Generate purchase orders based on reorder points and lead times
- Expiry Alerts: Generate reports of products approaching expiry
- Controlled Substance Tracking: Maintain audit trail with reporting capability
- Sales Recording: Record all product sales with automatic stock deduction
- Reporting: Generate reports for stock levels, expired products, low stock, controlled substance movements, and supplier performance

The system will be developed as a prototype PostgreSQL database-based solution, focusing on core pharmacy operations.

3.4 Boundaries

The Pharmacy Inventory System will not include:

- Medical Aid or Insurance Claims Processing: The system will capture medical aid details but will not submit claims electronically to medical aids
- Patient Consultation Records: Clinical notes beyond allergies and basic patient information are out of scope
- Point-of-Sale Hardware Integration: Sales will be recorded manually in the system, not integrated with cash registers
- Employee Payroll and HR: Staff management is limited to recording pharmacist IDs for dispensing logs
- E-commerce or Online Ordering: No online customer ordering functionality
- Detailed Financial Accounting: The system tracks cost and selling prices but does not generate financial statements (income statements, balance sheets)
- Integration with External Systems: No integration with supplier ordering portals or government health systems

3.5 Conclusion

This database system directly addresses major challenges in pharmacy inventory management, including stock inaccuracies, expiry tracking, and regulatory compliance. It clearly defines the required data, maintains a focused and manageable scope, and operates within academic and legal constraints.

References

BestRx (2023) 5 inventory management best practices for pharmacies. Available at: https://www.bestrx.com/blog/5-inventory-management-best-practices-for-pharmacies (Accessed: 28 March 2026).

CPCON Group (2026) Pharmacy inventory management: complete guide. Available at: https://cpcongroup.com/insights/article/pharmacy-inventory-management/ (Accessed: 28 March 2026).

Clicks Group (2025) Clicks Group profile and brands. Available at: https://www.clicksgroup.co.za/clicks-group-profile/brands/ (Accessed: 28 March 2026).

Gous, A., et al. (2020) Procurement challenges in Gauteng health services. BMC Health Services Research. Available at: https://doi.org/10.1186/s12913-020-05080-1 (Accessed: 28 March 2026).

Indus Journal of Bioscience Research (2025) The role of smart pharmacy automation in improving medication safety: a systematic review, 7(1), pp. 1628–1645.

Molope, R.E. (2020) Inventory management of chronic medicines. Master's thesis. University of Limpopo. Available at: http://ul.netd.ac.za/bitstream/handle/10386/3484/molope_re_2020.pdf (Accessed: 28 March 2026).

Nakambale, H., et al. (2023) Stockouts of medicines in KwaZulu-Natal. BMC Health Services Research. Available at: https://doi.org/10.1186/s12913-023-10096-4 (Accessed: 28 March 2026).

Oldland, A.R., Golightly, L.K., May, S.K., Barber, G.R. and Stolpman, N.M. (2015) 'Electronic inventory systems and barcode technology: impact on pharmacy technical accuracy and error liability', Hospital Pharmacy, 50(1), pp. 34–41.